Learnability, Reweighted: Which Tasks the Estimator Makes Active in Verifiable-Reward RL

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Abstract

RL with verifiable rewards often treats task curricula as independent of the group estimator that converts sampled outcomes into updates. We ask which tasks that estimator makes active. Every finite-group estimator induces an activity geometry over pass rates — the expected absolute mass of the coefficient vector it can emit — and for the practical MaxRL convention that drops all-fail groups this geometry has an exact closed form, $A_N(p) = 2(\text{pass@}N - \text{pass@}1)$, the estimator's own pass@N gap. It factors exactly as $p(1-p) \cdot w_{N-1}(p)$: the canonical learnability score reweighted by MaxRL's own objective weight at the truncation the deployed estimator targets, recovering $p(1-p)$ as the $N{=}2$ slice, with a peak that migrates toward harder tasks as $\log N/N$ and an all-fail rule that shifts the truncation order to $N-1$. This is an estimator-side diagnostic, not a theorem of learning progress, and we test it as a hypothesis. In a fresh 20-seed Acrobot tournament at $N{=}16$, the deployed-N score improves target-uniform AUC over $p(1{-}p)$ by $+.0480$ (95% paired-bootstrap CI $[+.0209, +.0738]$) and also beats uniform sampling in this fixed eight-threshold family; the effect replicates on two further platforms ($+.0322$, $+.0307$). Two preregistered follow-ups then bound the claim. Holding the deployed estimator at $N{=}16$ and sweeping only the score exponent, performance rises past $N{=}16$ (argmax at u_{64}), so the harder-peaked shape is what helps and the deployed-N peak location is not. Dropped into robust PLR on AMaze in place of MaxMC, the pure activity priority is starved of signal at one Bernoulli per level visit and does not beat the upstream baseline. A third, 48-block preregistration then bounds the task unit: where the curriculum scores a level aggregating heterogeneous mazes, the same shape contrast is negative ($-.0032$, CI $[-.0054, -.0011]$; an effect at or above the registered SESOI is ruled out). Theory explains it. The mass identity needs no independence at all — $A_N = 2(\Pr(K{>}0) - \mathbb{E}[K]/N)$ for any group law — so what fails is only the p-only reduction: scoring an aggregate by its mean pass rate over-predicts activity by exactly twice its excess all-fail probability, a penalty that grows with N and which the telemetry confirms to floating point. The controlled positives and these failures support a calibrated conclusion: finite-group coefficient activity is a principled, estimator-conditioned source of curriculum hypotheses, not a universal measure of learning utility — and mean pass rate is not a sufficient statistic for it once the curriculum names a coarser unit than the estimator consumes.

1 Introduction

RL with verifiable rewards (RLVR) converts automatically checkable outcomes into policy updates, and every update first pays for a group of N model rollouts. A task curriculum tries to make that budget productive by sampling tasks believed to be learnable now. The dominant curriculum score in the literature is learnability, $p(1-p)$ (Tzannetos et al., 2023; Rutherford et al., 2024): sample tasks the policy solves about half the time. We ask what

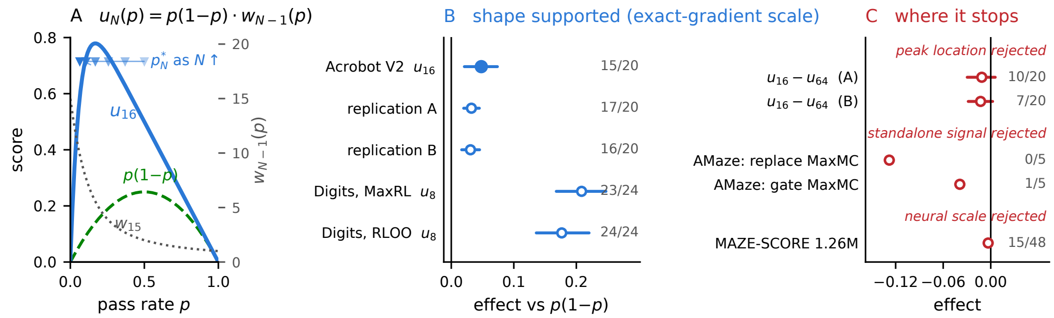

Figure 1: One identity and one claim boundary. (A) Activity is learnability reweighted by MaxRL's own objective weight (Cor. 1); the peak $p^\star_N$ is derived, not tuned. (B) Every measured contrast of the deployed-N shape against its $N{=}2$ slice, with paired-bootstrap 95% intervals and positive-pair counts; filled marks the one frozen confirmatory primary (§3.1). (C) The three preregistered boundaries. Rows here are different endpoints on their own studies' scales, shown together only to mark direction and interval; AMaze entries are five-seed development means at one-sixth budget and carry no interval, with a full-budget confirmatory campaign preregistered separately.

that score does not: learnable for which learner? Modern RLVR runs a group estimator — MaxRL (Tajwar et al., 2026), GRPO, RLOO — and the coefficients it can emit on a task are a function of that task's group outcome law and of N.

We answer for practical MaxRL. Its expected absolute coefficient mass obeys $A_N = 2(\Pr(K{>}0) - \mathbb{E}[K]/N)$ for any group law (Prop. 1); under conditionally i.i.d. rollouts this is $2\{1-(1-p)^N - p\} = 2(\text{pass@}N - \text{pass@}1)$ and factors as

$$u_N(p) = A_N(p)/2 = p(1-p) \cdot w_{N-1}(p),$$

where $w_T(p) = \sum_{k=1}^{T}(1-p)^{k-1}$ is MaxRL's own weight function (Cor. 1). Coefficient activity is learnability reweighted by the deployed estimator. $p(1-p)$ is the $N{=}2$ slice, and the peak $p^\star_N = 1 - N^{-1/(N-1)} \simeq \log N/N$ is derived, not tuned. MaxRL asked which objective more compute should buy; we ask which tasks that objective should be fed.

We then test the derived score as a curriculum hypothesis and report where it holds and where it stops:

1. The shape beats the learnability slice where gradients are exact. In a fixed Acrobot pool at deployed $N{=}16$, u_{16} beats $p(1-p)$ by +.048 under a frozen rule, replicated on two further platforms (+.032, +.031); an exact-probability Digits study shows the same shape effect at +.208 under MaxRL and +.177 under RLOO (Fig. 1).
2. The shape helps; the peak location does not. Holding the estimator at $N{=}16$ and sweeping the score exponent, performance rises past $N{=}16$ (§3.2). The value of u_N over $p(1-p)$ is in its tail — precisely where w_{N-1} differs from 1.
3. As a signal of its own it is starved. Dropped into robust PLR on AMaze in place of MaxMC, activity does not beat upstream: one Bernoulli per level visit cannot compete with a critic read at every timestep (§3.3).
4. And the p-only score does not survive a coarser task unit. In a 48-block preregistered test at 1.26M parameters and deployed $N{=}32$, where the curriculum scores a level that aggregates heterogeneous mazes, u_{32} loses to $p(1-p)$ (−.0032, CI [−.0054, −.0011]) — an effect at or above the registered SESOI is ruled out. Cor. 2 says why, and the telemetry confirms it to floating point (§3.4).
5. The estimator belongs in the curriculum's evaluation. Digits rejects a universal estimator-to-sampler mapping; the maze factorial shows the estimator ordering coverage under either sampler; a Countdown aggregate shows a logged proxy moving opposite to raw outcomes.

The identity is an estimator-side diagnostic, not a theorem of learning progress. The paper is deliberately narrower than the project record: exploratory mechanisms, failed hypotheses, robotics pilots, gate studies, and implementation detail remain in the appendix and artifact.

2 Coefficient activity under practical MaxRL

Setup. For a task x let $\mathbf{r} \in \{0,1\}^N$ be the group's rollout rewards, drawn from a joint law Q_x, and $K = \sum_i r_i$. Write $\bar{p} = \mathbb{E}_{Q_x}[K]/N$ and $q_N = \Pr_{Q_x}(K > 0)$; when the N rollouts are conditionally i.i.d. Bernoulli(p) these are p and pass@$N = 1 - (1-p)^N$. Let $s_i = \nabla_\theta \log \pi_\theta(y_i \mid x)$ be the response score and $\hat{g} = \sum_i w_i s_i$ the group update. Then pass@$N = 1 - (1-p)^N$. We analyze the zero-stabilizer idealization of the released practical centered MaxRL convention, which assigns

$$w_i = \frac{r_i}{K} - \frac{1}{N} \quad \text{when } K > 0, \qquad w_i = 0 \quad \text{when } K = 0.$$

All-pass groups also have zero contrast. The released implementation uses a finite denominator stabilizer ε, multiplying the ideal coefficients by $K/(K+N\varepsilon)$ when $K > 0$; the identities below are exact for the zero-ε convention, not for its finite-ε magnitude.

Definition 1 (Finite-group activity geometry). A binary group estimator $\mathcal{E}$ maps a group outcome $\mathbf{r}$ to coefficients $\mathbf{c}_\mathcal{E}(\mathbf{r}) \in \mathbb{R}^N$. Its activity geometry is the expected absolute mass it emits on a task whose group outcomes follow Q_x,

$$\mathcal{A}_{\mathcal{E},N}(Q_x) = \mathbb{E}_{\mathbf{r}\sim Q_x} \|\mathbf{c}_\mathcal{E}(\mathbf{r})\|_1 \overset{\text{symmetric}}{=} \sum_{k=0}^{N} Q_x(K{=}k)\, a_{\mathcal{E},k}, \qquad a_{\mathcal{E},k} = \|\mathbf{c}_\mathcal{E}(k)\|_1.$$

Under conditionally i.i.d. rollouts, $Q_x = \text{Bernoulli}(p_x)^{\otimes N}$ and this becomes the Bernstein polynomial $\sum_k \binom{N}{k} p_x^k (1-p_x)^{N-k} a_{\mathcal{E},k}$ — a one-dimensional geometry over pass rate.

Every estimator induces activity over a group outcome law; conditional i.i.d. sampling is what reduces that law to a geometry over a single scalar. Choosing an estimator fixes $\{a_{\mathcal{E},k}\}$ and therefore which tasks can emit an update at all, orthogonally to the objective-side question of which finite-rollout objective to deploy (Tajwar et al., 2026; Zheng, 2026). What singles out practical MaxRL is that its mass is affine in the two statistics $(q_N, \bar{p})$, so the sum collapses without any independence assumption.

Lemma 1 (Practical truncation order). The expectation of the drop-$K = 0$ practical estimator equals the gradient of the $(N-1)$-term truncated maximum-likelihood objective, not the N-term objective.

For $J_T(p) = -\sum_{t=1}^{T}(1-p)^t/t$, direct conditioning gives

$$\mathbb{E}\hat{g} = \left(\frac{1-(1-p)^N}{p} - (1-p)^{N-1}\right)\nabla p = \left(\sum_{j=0}^{N-2}(1-p)^j\right)\nabla p = \nabla J_{N-1}.$$

The correction matters whenever a theoretical truncation parameter is mapped to a deployed group size. This $N-1$ statement does not revise MaxRL's theorem for its uncentered conditional estimator; it characterizes the released centered convention after its all-fail group is zeroed, so the shift is a property of the deployed rule, not of the maximum-likelihood expansion. Details and the non-dropped control-variate convention are in App. A.

Proposition 1 (Exact coefficient mass under any group law). For any joint binary group law Q — no independence and no identical distribution required — the practical estimator's expected absolute coefficient mass is

$$A_N(Q) = \mathbb{E}_Q\left[\sum_{i=1}^{N} |w_i|\right] = 2\,(q_N - \bar{p})\,.$$

If the rollouts are conditionally i.i.d. Bernoulli(p) this is

$$A_N(p) = 2\{1 - p - (1-p)^N\} = 2(\text{pass@}N - \text{pass@}1).$$

The realized mass of a group is the deterministic function $M(K) = 2(1 - K/N)\mathbf{1}\{K > 0\}$ — successes and failures each contribute $(N - K)/N$ when $0 < K < N$ — so $A_N(Q) = \mathbb{E}_Q[M(K)] = 2(q_N - \bar{p})$ needs no distributional assumption at all; only the scalarization to p does. This identity concerns coefficients, not gradient norm, signal-to-noise ratio, or expected improvement: score-gradient geometry can change its downstream effect.

The bridge to the expected update is exact: with $\mu_\pm = \mathbb{E}[s_i \mid r_i{=}1/0, x]$ we have $\mathbb{E}[\hat{g} \mid x] = u_N(p)(\mu_+ - \mu_-)$ for $u_N = A_N/2$, so u_N is the estimator-controlled scalar factor in the expected update and the conditional score geometry stays outside the mass calculation. Under i.i.d. rollouts $u_N(p) = \Pr(r_1{=}0,\, K > 0) = (1-p)\{1 - (1-p)^{N-1}\}$: activity is the probability that one rollout fails while a peer in the purchased group succeeds. Both dead zones follow — an all-fail task supplies no positive contrast, an all-success task no negative contrast — as does why N belongs in the curriculum's definition rather than its implementation.

Corollary 1 (Activity is reweighted learnability). Write MaxRL's population gradient as $\nabla J_T = \mathbb{E}_x[w_T(p)\nabla p]$ with weight function $w_T(p) = \sum_{k=1}^{T}(1-p)^{k-1}$ (Tajwar et al., 2026). Then, exactly,

$$u_N(p) \;=\; p(1-p)\cdot w_{N-1}(p).$$

By Lemma 1 the deployed estimator targets $T = N - 1$, and $p(1-p)\sum_{k=1}^{N-1}(1-p)^{k-1} = (1-p)\{1-(1-p)^{N-1}\} = u_N(p)$. So coefficient activity is the learnability score reweighted by MaxRL's own weight function at the truncation the deployed estimator targets: $p(1-p)$ peaks at $\frac{1}{2}$ regardless of estimator, w_{N-1} decreases in p and grows with N as $p \to 0$, and their product peaks at $p_N^\star = 1 - N^{-1/(N-1)}$, where an informative rollout meets an estimator that still cares. The canonical binary-success score $p(1-p)$ — used by SFL and LILO (Rutherford et al., 2024; Foster et al., 2025) and recovered here as RLOO's coefficient mass — is therefore the $N{=}2$ slice: it scores tasks as if the learner were REINFORCE with one rollout, while u_N scores them for the estimator deployed.

Corollary 2 (Task granularity). Suppose a curriculum scores an aggregate z (a level, template, or domain) whose concrete tasks are $X \sim \nu(\cdot \mid z)$ with conditionally i.i.d. rollouts given X. Because $u_N''(p) = -N(N-1)(1-p)^{N-2} < 0$, plugging the aggregate's mean pass rate $\bar{p}_z$ into u_N over-predicts its activity, and the gap is exactly the aggregate's excess all-fail probability:

$$A_N(\bar{p}_z) - 2\,\mathbb{E}_X[u_N(p_X)] = 2\left[\Pr(K{=}0 \mid z) - (1-\bar{p}_z)^N\right] \;\geq\; 0.$$

Both sides follow from Prop. 1: the aggregate's own group law has $q_N = 1 - \Pr(K{=}0 \mid z)$, so $2(q_N - \bar{p}_z)$ is its true activity and subtracting it from $A_N(\bar{p}_z)$ leaves the excess-silence term. Curvature says who pays: near $p_N^\star$ the penalty is $\approx \frac{1}{2}|u_N''|\operatorname{Var}(p_X)$ with $|u_N''(p_N^\star)| = (N-1)/(1-p_N^\star) \approx 34.7$ at $N{=}32$, against the exact and far milder $u_2(\bar{p}) - \mathbb{E}[u_2(p_X)] = \operatorname{Var}(p_X)$. Raising N moves activity toward harder tasks and makes the geometry more sensitive to how coarsely the curriculum names a task (§3.4).

Across estimators. As $p \to 0$, MaxRL carries $(N-1)$ times the coefficient mass of standard $1/N$-averaged RLOO; as $p \to 1$ both decay at first order in $1-p$, whereas sample-SD-normalized GRPO — whose Bernstein sum does not collapse — allocates relatively more mass to near-mastered prompts (Fig. 3, App. A).

Fixed-completion check. On a 36-task CPU skill-chain testbed at fixed sampled-completion budget, the mean normalized pass-rate AUC difference $u_N - p(1-p)$ rises .0000, +.0307, +.0920, +.1526, +.1909 over $N \in \{2, 4, 8, 16, 32\}$, positive in all eight paired seeds for every $N > 2$, though only 5/8 trajectories are monotone and u_N passes uniform only at $N \geq 8$. Descriptive, not preregistered; protocol and the uniform contrasts are in App. C.

Exact-probability counter-test. The stronger test does not generalize the sampler mapping. Under an internally hashed local protocol without an immutable public pre-execution commit, a fresh 24-block Digits contextual bandit computes the policy's exact correct-class probability before every update. Its frozen estimator-by-sampler interaction, +.01589 (95%

CI $[-.01686, +.04712]$, $p = .350$; 15/24), is not supported: MaxRL favors u_8 over $p(1-p)$ by +.20842 (23/24), but so does RLOO (+.17665, 24/24), reversing its prediction, and both matched samplers fall below uniform. One study thus carries the paper's largest shape effect and its sharpest negative.

Instantiation. FrontierMax operationalizes the geometry without changing the within-group estimator: each task keeps discounted Beta evidence, draws a posterior pass rate $\tilde{p}_x$, and is sampled with probability $q(x) = \epsilon/M + (1-\epsilon)[u_N(\tilde{p}_x)]^\gamma / \sum_j [u_N(\tilde{p}_j)]^\gamma$. The floor ϵ, concentration γ, decay, and posterior estimator remain empirical choices; the theory supplies the utility and its N-dependence, not a parameter-free scheduler. An exact deterministic posterior priority is also available — for $p_x \mid \mathcal{D} \sim \mathrm{Beta}(a_x, b_x)$ with $(z)_N$ the rising factorial, $\mathbb{E}[u_N(p_x) \mid \mathcal{D}] = 1 - (b_x)_N/(a_x+b_x)_N - a_x/(a_x+b_x)$, which the AMaze arm uses; scoring the posterior mean instead overstates activity by a closed-form uncertainty penalty (App. A), the same concavity that drives Cor. 2. Coefficient activity is the shared quantity, not a license to conflate a proved identity, a tested fixed-pool teacher, and an untested replay overlay; Table 2 in App. B attaches a status to each object separately.

3 Evidence

The Acrobot tournament isolates the deployed-N score shape under a fixed scaffold where the scored unit is the task itself; the maze factorial tests an estimator-conditioned coverage ordering, not the score; and Digits supplies both the boundary on estimator-matching and the largest shape effect here. The positive support is deliberately controlled and, as §3.4 shows, measurably bounded. Figure 1 states for each contrast what it is and is not: one frozen primary, concordant supporting measurements, three preregistered boundaries.

3.1 Fresh fixed-pool Acrobot score test

We isolate the arbitrary-N score on official Gymnasium Acrobot-v1, where the scored unit is the task. Eight nested tip-height predicates share one task-blind H64 actor (640 parameters), the hardest being exactly native Acrobot success. All arms use practical MaxRL at $N = 16$, no hindsight, learning rate 3×10^{-4}, and a nominal two million paid transitions; twenty fresh paired seeds compare uniform sampling, $p(1-p) = u_2$, and u_{16} under identical discounted Beta tracking. The internally frozen primary is target-uniform normalized transition-AUC for $u_{16} - p(1-p)$, requiring a positive exact two-sided sign-flip test at .05 and a point estimate of at least +.01.

Table 1: Fresh Acrobot tournament. Paired differences use 20 fresh seeds under the source/runtime-locked protocol. The two uniform comparisons form one Holm family.

contrast	mean Δ	95% boot. CI	exact p	Holm p	decision
$u_{16} - p(1-p)$	+.0480	[+.0209,+.0738]	.0034	–	supported by rule
$p(1-p)$−uniform	-.0062	[-.0226,+.0122]	.5078	.5078	not supported
u_{16}−uniform	+.0419	[+.0218,+.0606]	.0008	.0016	supported

Arm means are .64523 (uniform), .63907 ($p(1-p)$), .68711 (u_{16}). The primary clears both frozen criteria with 15/20 positive pairs, supporting the deployed-N shape against its N=2 ablation in this fixed family — while $p(1-p)$ itself is not supported over uniform by the frozen secondary.

3.2 Does the deployed-N peak location matter?

The peak $p^\star_N = 1 - N^{-1/(N-1)}$ falls $.500 \to .169 \to .064$ for $N = 2, 16, 64$, so in every study above the winning arm is also the harder-peaked arm: "harder peaks beat $p(1-p)$" and "the deployed-N peak is correct" predict identically there, and only the second would advance on the $p(1-p)$ literature. Two preregistered campaigns separate them, holding the deployed estimator at $N = 16$ in every arm and varying only the sampling exponent, on the same 20 logical seeds and frozen scaffold, analyzed once each.

An over-shooting arm scored by u_{64} does not lose to u_{16}: the paired difference $u_{16} - u_{64}$ is $-.0113$ (95% CI $[-.0297, +.0057]$, exact $p = .235$, 10/20) on one platform and $-.0128$ ($[-.0284, +.0024]$, $p = .131$, 7/20) on a second. A seven-point exponent sweep $N_{\text{score}} \in \{2, 4, 8, 16, 32, 64, 128\}$ gives arm means .644, .668, .665, .675, .679, .688, .681, with argmax at u_{64}, not at the deployed u_{16}, and Spearman rank correlation +.93 between exponent and mean. The curve rises past the deployed N: peak-location specificity is not supported at this operating point.

The same campaigns replicate the shape effect. Against $p(1-p)$, u_{16} wins by $+.0322$ ($[+.0198, +.0449]$, exact $p = .000105$, 17/20) and $+.0307$ ($[+.0166, +.0453]$, $p = .000507$, 16/20), versus $+.0480$ in V2 — the first reproductions outside the machine that produced V2, both with a smaller p-value than V2's own. Magnitude is platform-sensitive (about 35% attenuation on identical seeds between architectures), so we attribute each number to its platform. Because the two new campaigns share seeds and a deterministic engine, their agreement is a portability check, not additional independent seeds.

Why the curve rises past the deployed N. Branching an exact-gradient pool to measure true continuation utility $U_H(x) = J(\text{Train}_H(\theta; x)) - J(\theta)$ at $H = 8$, u_{16} ranks it far better than $p(1-p)$ (+.106, 10 seeds, exact $p = .002$), while under RLOO — whose mass is $2p(1-p)$ — the two are indistinguishable (+.0001). But U_H peaks near $p \approx .02$, below both $p^\star_{16}$ and $p^\star_{64}$: activity ranks utility well and locates it poorly. Local activity and downstream benefit are also reported to diverge across domains under GRPO (Yang et al., 2026). Mechanism on 36 synthetic tasks, not evidence for a sampler.

The contrast stays descriptively positive on sampled-group (+.05294) and optimizer-update (+.05366) axes, and u_{16} receives 2.40 fewer groups and 3.42 fewer updates per million transitions, so it did not win by receiving more updates; these non-confirmatory axes identify no causal mechanism. Native-success transition-AUC is .31085, .30521, .37615 across arms. App. C reports the near-equal observed mass means and the audited but RNG-confounded precursor.

3.3 Where the score is starved: activity as a PLR priority on AMaze

The natural competitive test drops the activity priority into minimax's robust PLR (Jiang et al., 2021; 2023) in place of MaxMC, running the shipped maze configurations verbatim. Our development sweep — seeds disjoint from any confirmatory block, 5,000 of the 30,000 shipped updates, five paired seeds per arm, held-out return over the three shipped test mazes — is negative, and the reason is informative. Upstream robust PLR scores .629 and the no-curriculum control .539, with every activity variant below the control: replacing MaxMC with $\mathbb{E}[u_N(p) \mid \mathcal{D}]$ at the shipped 32×1 batch scores .500 (N=8; 0/5 seeds beat MaxMC, exact p=.0625) and .551 (N=16), while a 4×8 regrouping that buys evaluation fidelity with level diversity costs $-.207$ for MaxMC too.

The cause is information, not shape. MaxMC reads critic disagreement $\max\text{-return} - V(s_t)$ at every timestep, with memory of the best return ever achieved on that level, while at the shipped n_eval=1 a success-rate score gets one terminal Bernoulli per level visit — a two-valued posterior, nearly all ties under rank prioritisation — and max_episode_steps=250 against a 256-step rollout forbids a second episode. Gating MaxMC by activity rather than replacing it recovers most of the gap (.590; $-.039$ to upstream, 1/5) without clearing it. The AMaze student is PPO with GAE and never forms binary rollout groups, so the faithful mapping is outside this evidence.

3.4 A coarser task unit: the p-only score does not transfer

Every positive above is at 640 or 4,810 parameters with exact or near-exact gradients. We preregistered the same contrast at neural scale: a 1.26M-parameter transformer on procedural 17×17 mazes across 13 goal-distance levels, estimator fixed to practical MaxRL at the deployed $N = 32$, three samplers (u_{32}, $p(1-p)$, uniform) sharing one Beta posterior, decay, floor, level set, and per-block SFT checkpoint. Forty-eight paired blocks; endpoint

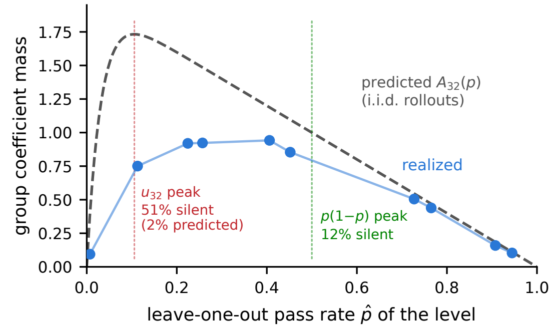

Figure 2: The task-granularity gap (descriptive, computed after the frozen primary). Plug-in prediction evaluates A_{32} at a level's mean pass rate, estimated leave-one-out from the other groups on that level so mass is never binned against its own K. Realized mass tracks the predicted shape ($r = .90$, 288,000 group draws) but lies below it by exactly twice the level's excess all-fail probability (Cor. 2), a gap that widens as p falls.

mean pass@8 coverage over ten RL evaluations minus its post-SFT value; support required Holm $p < .05$, a positive bootstrap lower bound, and at least $+.005$.

The primary is practically ruled out by the frozen rule: u_{32} minus $p(1-p)$ is $-.00324$ (95% CI $[-.00543, -.00111]$, exact $p = .0054$, 15/48), the interval's upper bound falling below the SESOI. The registered secondary is supported (u_{32} beats uniform by $+.00888$, $[+.00657, +.01115]$, 41/48), so curriculum sampling helps here; what fails is that the rollout-aware shape is the better one. Per the branch fixed in advance, effects at or above the SESOI are ruled out under this protocol and no universal non-transfer is claimed.

Why: the curriculum names a coarser unit than the theory. The trainer draws one concrete maze per group and repeats that prompt N times, while the teacher's posterior pools over the many mazes of a level. The conditionally i.i.d. unit is the maze; the unit scored is the level. So Prop. 1 still holds exactly — what fails is its p-only reduction, and Cor. 2 prices the failure. The telemetry confirms both identities to floating point: across 41,101 (seed, arm, level, window) cells realized mass equals $2(\hat{q}-\hat{p})$ and the plug-in over-prediction equals twice the excess all-fail probability, each to $< 5\times10^{-16}$, unchanged at window widths 10, 25, 50 (App. C). The gap is large where the score aims: at $\hat{p} \approx .11$ the binomial predicts 2.2% silent groups and 51.2% are silent (Fig. 2). Clustered on the 48 seed blocks, u_{32} realizes .580 $[.570, .590]$ of its prediction against .703 $[.691, .715]$ for $p(1-p)$ (paired $-.123$, 48/48 blocks negative), losing 60.8% of its groups to silence against 32.1% — as Cor. 2 predicts for the harder-peaked score. This exactly accounts for the mass prediction error and is consistent with the endpoint, but it is not an intervention and we claim no sole mediation. Post-hoc; no registered quantity changes.

Externally recorded maze factorial (secondary). A balanced six-block factorial crossing $\{\text{MaxRL}, \text{GRPO}\} \times \{\text{uniform}, \text{FrontierMax}\}$ on the same maze learner tests an estimator ordering, not the score. Its first wave rejected the stronger externally recorded endpoint claim (3/6, 4/6) and we retracted the earlier zero-exception cohort claim; the re-registered wave-2 primary, time-integrated pass@8 coverage, is positive in all six fresh blocks under each sampler (exact sign $p = .03125$ per sampler; block mean $+.0195$, $[+.0115, +.0275]$) — an estimator-conditioned coverage ordering at common optimizer settings, not universal estimator superiority. Protocol and retraction record in App. C.

Secondary records. Paid-probe selection, failure recycling, and the historical Countdown aggregate are reported only in App. C and support no named-method superiority claim; their measurement lesson is that raw per-task outcomes are needed to distinguish mean accuracy from standard pass@k.

4 Related work

Objective-side work such as MaxRL (Tajwar et al., 2026) and RL2ML (Zheng, 2026) characterizes which population or finite-rollout objective a group estimator implements, typically holding the task distribution fixed. Learnability curricula such as ProCuRL (Tzannetos et al., 2023), SFL (Rutherford et al., 2024) and LILO (Foster et al., 2025) adapt that distribution using success variability, while advantage-bandit curricula such as SEC (Chen et al., 2025) and DUMP (Wang et al., 2025) estimate activity empirically from observed policy advantages, and TAC (Yang et al., 2026) augments local activity with first-order cross-domain transferability under GRPO. We study a complementary object: the exact task-level finite-group activity geometry a deployed estimator induces (Def. 1), derived from its algebra before any curriculum hypothesis is tested, with a peak that migrates with the rollout budget and used for task selection rather than objective redesign. Superiority over the methods below is untested and not claimed.

Concurrent exact analyses of group estimators. A concurrent cluster derives closed-form expected update magnitudes on the standard-deviation-normalized branch: the group-SD identity with DAPO's silent-group mass $p^G + (1-p)^G$ (Bay, 2026), an expected absolute GRPO advantage of $2\sqrt{p(1-p)}$ motivating 50%-success targets in a learned-curator bandit (Gu et al., 2026), and energy- or variance-based selection scores (Cui, 2026). There the magnitude peaks at $p = .5$ independent of N; the practical-MaxRL mass is the estimator's own unnormalized pass@N gap, peaking as $\ln N/N$. MoPPS (Qu et al., 2026) anticipates the posterior machinery; our novelty claim is only the utility sampled toward.

Curricula and rollout allocation. Intermediate-difficulty sampling predates this work: under ProCuRL's assumption that optimal-policy success is one its proximal score reduces to $p(1-p)$ (Tzannetos et al., 2023), and SFL (Rutherford et al., 2024) and LILO (Foster et al., 2025) prioritize binary reward variance. Our distinction is convention-specific: $u_N = A_N/2$ is the arbitrary-N expected half-mass of the deployed centered MaxRL rule, derived from its algebra rather than estimated online. VCRL (Jiang et al., 2025) thresholds realized group variance; PLR (Jiang et al., 2021) and ALP-GMM (Portelas et al., 2020) use TD error, staleness, or learning progress, and so react to forgetting where instantaneous mass cannot — our AMaze arm is robust PLR (Jiang et al., 2023) with only the replay priority replaced (§3.3). Generator-based UED — PAIRED (Dennis et al., 2020), ACCEL (Parker-Holder et al., 2022) — creates tasks rather than pricing a fixed pool. GRESO (Zheng et al., 2025) and DPS (Mao et al., 2026) select prompts from reward dynamics before rollout, VIP (Nguyen et al., 2026) varies per-prompt rollout counts to cut predicted gradient variance, and allocators stage extra rollouts toward high-value prompts (HORA (Wang et al., 2026), VIGOR (Jiang et al., 2026), DARS (Yang et al., 2025), Reinforce-Ada (Xiong et al., 2025)); we instead select the task distribution before generation at fixed N. HORA's posterior-integrated hit utility makes the move Cor. 2 formalizes — integrate a nonlinear rollout utility over the relevant distribution rather than evaluate it at a mean — but for epistemic uncertainty about one prompt, where ours is heterogeneity inside a scored aggregate. BBG (Yuan et al., 2026) gates updates by posterior marginal pass@k contribution and boundary-aware Curriculum RL (Cai et al., 2026) adds guidance near a reasoning boundary. These target hit probability, variance, exploration, boundary metrics, or signal creation — not practical-MaxRL coefficient mass. RL2ML (Zheng, 2026) separates finite-rollout objectives, update scale, variance, and metric gain; its $T = N$ alignment retains $K = 0$ via the direct estimator or a score-baseline control variate, while our $T = N - 1$ statement is for the deployed hybrid that centers only when $K > 0$.

Coverage measurement. RLVR can improve pass@1 while reducing high-k coverage (Yue et al., 2025; Yuan et al., 2026; Zhou et al., 2026); entropy collapse is complementary evidence (Cui et al., 2025) and pass@k objectives address the metric directly (Chen et al., 2025). We ask whether an external curriculum changes that coverage under two estimators. The secondary hindsight and relabeling record (App. C) is not evidence for the acquisition score, but it marks the limit of allocation: a sampler redistributes existing signal, while relabeling or subproblem decomposition (Jiang et al., 2026) can create signal outside its support — at $A_N = 0$ no priority helps.

5 Limitations

Coefficient mass ignores score-gradient direction, parameter sharing, optimizer state, task transfer, and non-i.i.d. rollouts: it is a sampler-facing surrogate, not a guarantee of loss reduction or metric improvement. The fixed-completion N-sweep is post-guidance (designed after the theoretical guidance existed, hence not preregistered) on one co-designed synthetic chain, and the exact-probability Digits counter-test is a contextual bandit rather than trajectory RL. The fresh V2 Acrobot score test uses one fixed eight-threshold family, one 640-parameter learner, and no held-out task distribution or native named-method implementation; its 20 seeds were inherited from a historical precursor rather than set by prospective power analysis, and its exact paired test requires sign exchangeability under the sharp null. V2 paired no mismatched-N score against a fixed deployed N; §3.2 does, and finds no peak-location specificity, so the evidence supports rollout-aware difficulty targeting and specifically not the deployed-N peak. The four evidence families froze their primaries independently at $\alpha = .05$; we claim no paper-level error rate. Retained evaluation records do not include per-episode trajectories. The maze factorial uses a small transformer, one task family, and common rather than estimator-tuned learning rates; its easy-band mechanism remains descriptive. MAZE-SCORE (§3.4) rules out effects at or above its $+.005$ SESOI for one task family, one architecture, one N, and one budget. Its protocol varies model capacity, function approximation, task heterogeneity and curriculum granularity together, so "scale" is not an isolated causal variable here. The granularity diagnosis is post-hoc and exactly accounts for the coefficient-mass prediction error, but no intervention establishes it as the sole mediator of the endpoint; its windowed estimates also pool maze-to-maze heterogeneity with learner nonstationarity, which the window-invariance of the identity bounds but does not remove. Whether scoring $\hat{q}_z - \hat{p}_z$ directly, or a task-level posterior, closes the gap is untested here and we assert no counterfactual outcome. The Coefficient-Activity PLR row in Table 2 is a development negative (§3.3), not preregistered evidence, and we claim no advantage over PLR, robust PLR, ACCEL, or any other minimax baseline. No robotics performance is reported; the paid-probe, recycling, Countdown, gate, and transfer records are secondary and confined to App. C.

Registration timing is the recurring gap. The Digits lock and the Acrobot tournament lock are internally hashed — the latter binds source, runtime, gates, and artifacts — but neither is backed by an immutable public pre-execution commit independently establishing timing, and the external maze wave-2 record, though specified before execution, has no locking commit object vendored here: the release can audit stored protocols and verdicts, not their timing. Artifact coverage is likewise incomplete, with maze checkpoint trajectories external and the Digits replay payload local without a download URI; App. D itemizes these gaps alongside the 53-row compact registry and the distinct release-branch audit object. Together they limit auditing and preclude a broad neural claim.

6 Conclusion

The estimator belongs in the evaluation of data-selection design: its algebra fixes which tasks can emit activity at a given rollout count. In a fixed classical-control family u_{16} beat both u_2 at the deployed N=16 and uniform sampling, twice replicated; Digits rejects a universal mapping. Three preregistered follow-ups drew the boundary: performance rises past the deployed N, so the shape helps and its peak location does not; as a standalone PLR priority activity is starved at one Bernoulli per level visit; and where the curriculum scores a level aggregating heterogeneous mazes the contrast reverses, by exactly the excess all-fail probability of Cor. 2. The estimator defines the coefficient map; the curriculum defines the unit over which it is averaged, and the two do not commute. Activity diagnoses available update, not learning utility, and a mean pass rate determines it only when the scored unit is the estimator's own. Evaluate data selection with the estimator it ships on, report pass@k from retained outcomes, and treat activity as a gate, not a law.

Reproducibility statement

The repository includes exact-enumeration tests, a one-command artifact check, structured seed-block analyses, stored protocol and verdict documents, figure derivation scripts, a source-locked 60-run Acrobot tournament with deterministic retained-ledger reanalysis, and a generated 53-row compact registry for locally indexed maze, Countdown, and GSM8K artifacts. Secondary records and receipt manifests remain documented in the appendix. Registration timing objects that remain external are disclosed rather than treated as locally auditable. The independent unit is stated for every central result; known raw-data gaps and secondary-study provenance are disclosed in App. D.

AI use statement

AI assistants supported planning, implementation, debugging, literature discovery, statistical and artifact auditing, figure preparation, and manuscript editing. The authors selected the questions and decision rules, ran the studies, verified derivations, code, numerical results, and citations, and take responsibility for every claim.

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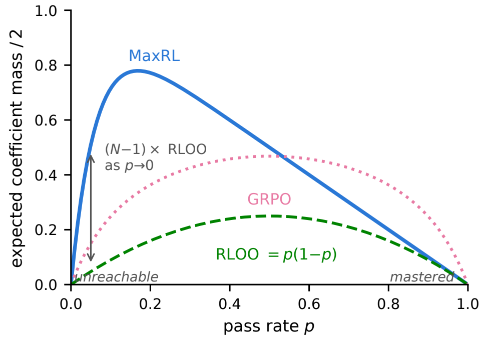

Figure 3: Three estimators, three activity geometries. Exact expected half-mass at $N = 16$ under zero-stabilizer MaxRL, gradient-averaged RLOO, and the zero-stabilizer sample-SD-normalized GRPO convention. Coefficient mass is sign-blind, so the curves motivate rather than prove the empirical coverage ordering.

A Proof details

Truncation order. For $K > 0$, conditioning on one sampled response and using exchangeability gives

$$\mathbb{E}\hat{g} = \left(\frac{1-(1-p)^N}{p} - (1-p)^{N-1}\right)\nabla p.$$

Since $1-(1-p)^N = p\sum_{j=0}^{N-1}(1-p)^j$, subtracting the final term leaves $\sum_{j=0}^{N-2}(1-p)^j\nabla p = \nabla J_{N-1}$. If the centered control-variate term is retained on $K = 0$, its contribution is $(1-p)^{N-1}\nabla p$. The expectation becomes $\sum_{j=0}^{N-1}(1-p)^j\nabla p = \nabla J_N$. Thus the direct/non-dropped convention has order N, while the deployed hybrid convention above has order $N-1$; theorem and implementation must use the same convention. At audited MaxRL source commit 7197bbb, the released path computes $(r_i - \bar{r})/(\bar{r}+\epsilon)$ and zeros all-failure groups; after the deployed gradient average, its $\epsilon \to 0$ coefficients match this convention up to common scale. The parsed truncation-order setting is inactive on that path.

Coefficient mass. For $0 < K < N$, each success has coefficient $(N-K)/(KN)$ and each failure has coefficient $-1/N$. The two classes each contribute $(N-K)/N$ in absolute value. For $K \in \{0, N\}$ the deployed centered estimator contributes zero. With $K \sim \mathrm{Bin}(N, p)$,

$$\begin{aligned}
A_N(p) &= \frac{2}{N}\sum_{k=1}^{N-1}(N-k)\binom{N}{k}p^k(1-p)^{N-k} \\
&= \frac{2}{N}\{\mathbb{E}[N-K] - N\Pr[K=0]\} \\
&= 2\{1-p-(1-p)^N\}.
\end{aligned}$$

The peak follows from $u_N'(p) = N(1-p)^{N-1} - 1$ and strict concavity for $N > 1$.

Posterior-integrated activity. For $p \sim \mathrm{Beta}(a, b)$, $\mathbb{E}[p] = a/(a+b)$ and the beta-function ratio gives

$$\mathbb{E}[(1-p)^N] = \frac{B(a, b+N)}{B(a,b)} = \frac{(b)_N}{(a+b)_N}.$$

Substitution in $u_N(p) = 1-(1-p)^N - p$ proves the deterministic posterior priority in §2. Subtracting it from $u_N(\mathbb{E}[p])$ gives the printed nonnegative Jensen gap.

Estimator limits. Under the deployed $1/N$ gradient-average normalization, RLOO expected half-mass is $p(1-p) = u_2(p)$, so $u_N(p)/u_2(p) \to N-1$ as $p \to 0$. Near mastery, both are first order in $q = 1-p$. With sample-SD normalization, GRPO group mass

Table 2: Contribution–evidence map. Status is attached to the specific object evaluated, not transferred across rows.

Object	Claim/mechanism	Evidence allowed here
Coefficient identity	Exact A_N, $T = N - 1$, and posterior-integrated u_N for the stated practical-MaxRL convention	Proved; estimator activity only, not learning progress
Direct fixed-pool teacher	Thompson-scored u_N with proportional sampling and a uniform floor	Scoped empirical; Acrobot supports the score in one fixed family, while Digits rejects a universal mapping
Coefficient-Activity PLR	Analytic posterior activity replaces MaxMC priority inside robust PLR; replay and staleness remain PLR mechanisms	Development negative (§3.3); as a replacement for MaxMC it does not beat upstream robust PLR on AMaze at 5,000 updates

is proportional to $\mathbb{E}[\sqrt{K(N-K)}]$ up to deployed normalization constants; its relative tail scaling is verified by exact enumeration in verify_guidance_math.json. These are coefficient statements, not gradient-direction statements.

Dependent rollouts. The per-group identity does not require independence. For any binary group,

$$\mathbb{E}\sum_i |w_i| = 2\{\Pr(K \geq 1) - \mathbb{E}[K]/N\}.$$

With a common marginal p, the second term is p; independence is used only to replace $\Pr(K \geq 1)$ by $1-(1-p)^N$. In a post-guidance exact beta-binomial CPU stress test, count-distribution enumeration matched the general identity to 2.58×10^{-13}. At $\rho = .10$, the activity peak moved from .169 to .234 for $N = 16$ and from .106 to .186 for $N = 32$. This dependence model shows how increasing positive correlation can lower activity inside that family; pairwise correlation alone does not determine a general joint law. The result is a scope diagnostic, not evidence that coefficient mass causes learning or that beta-binomial dependence describes neural rollouts.

Posterior-mean bias and the uncertainty penalty. With $p_x \mid \mathcal{D} \sim \mathrm{Beta}(a_x, b_x)$, strict concavity of u_N for $N > 1$ and Jensen's inequality give $\mathbb{E}[u_N(p_x) \mid \mathcal{D}] \leq u_N(\mathbb{E}[p_x \mid \mathcal{D}])$, so a priority computed from the posterior mean overstates expected activity. The gap is nonnegative and closed-form,

$$u_N\left(\frac{a_x}{a_x+b_x}\right) - \mathbb{E}[u_N(p_x) \mid \mathcal{D}] = \frac{(b_x)_N}{(a_x+b_x)_N} - \left(\frac{b_x}{a_x+b_x}\right)^N,$$

and vanishes as the posterior concentrates. It is an uncertainty penalty, not an exploration bonus: it shrinks the priority of tasks whose pass rate is still poorly determined.

B Algorithms and estimator contracts

Posterior teacher. Each task stores discounted successful and failed rollout counts. A Thompson draw $\tilde{p}_x$ is mapped through u_N and mixed with a uniform floor. Teacher state, sampler cursor, and checkpoints restore together. Observations use pre-relabel rewards so recycled outcomes do not inflate the requested task's posterior.

Coefficient-Activity PLR (development negative). The AMaze implementation is a narrow overlay on robust PLR in the pinned minimax substrate (Jiang et al., 2021; 2023). It uses cumulative Beta evidence per replay-buffer slot and replaces MaxMC priority with deterministic $\mathbb{E}[u_N(p) \mid \mathcal{D}]$; the declared N may equal n_eval (exact grouping) or be decoupled from it. Buffer admission, the robust replay gate, inverse-rank mass, staleness mixing, and with-replacement draws remain PLR mechanisms. Tie-aware ranking is shared with the matched MaxMC control. §3.3 reports the development sweep: the replacement priority does not beat upstream robust PLR, and no baseline-performance claim is made.

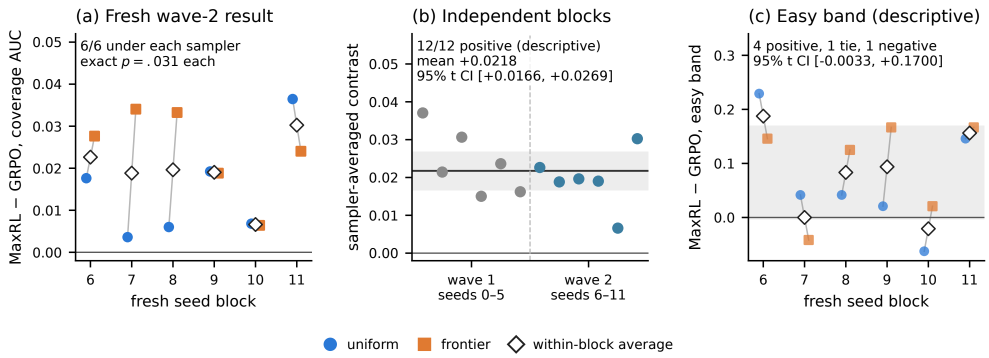

Figure 4: Maze factorial at the independent seed-block unit. (a) The externally recorded fresh-wave result is positive in 6/6 blocks under each sampler (exact sign $p = .03125$ each). Connected observations share a block. (b) Sampler-averaging gives wave-2 mean $+.0195$ and post-hoc 95% CI $[+.0115, +.0275]$; 12/12 cross-wave block averages are descriptive. (c) Easy-band localization is suggestive: four positive blocks, one tie, one negative, and an interval containing zero.

Relabeling (secondary). For an all-fail request, a verified achieved goal can create an auxiliary update where repeating the request would emit none. A parser proposes that goal from a failed response, and the environment's exact verifier must accept the response under that goal. Prompt conditioning and explicit response goal declarations are rewritten; answer spans, arithmetic, decimals, signed values, and unknown contexts are preserved. The reported-run implementation is frozen, while a conservative release wrapper and 26 CPU regression/property tests protect future runs. Relabeled groups remain auxiliary selected-data updates; no importance ratio converts them into fresh on-policy target samples. The intervention changes both update dose and direction, and its achieved-goal distribution is biased toward outputs the policy already reaches (Andrychowicz et al., 2017; Ghosh et al., 2021).

C Experimental protocols and secondary results

Fixed-completion N-sweep (post-guidance, descriptive). 36-task CPU skill-chain testbed, $N \in \{2, 4, 8, 16, 32\}$, eight paired seeds, no recycling, exactly 51,200 sampled completions per cell, teacher decay normalized per completion so changing N does not change the memory horizon. Against uniform the contrasts are $-.0106$, $-.0032$, $+.0309$, $+.0453$, $+.0836$; positive-seed counts against $p(1-p)$ are identical at $N{=}2$ then 8/8, 8/8, 8/8, 8/8, and against uniform 3/8, 3/8, 8/8, 7/8, 7/8. The $N{=}2$ identity reproduces bit-for-bit. Designed after the theoretical guidance existed, hence not preregistered, and on one co-designed testbed: evidence for rollout awareness, not a scaling law or a blanket curriculum advantage.

Fixed-completion N-sweep. The environment contains three nested 12-level chains and ten actions. Every cell runs exactly 51,200 completions, with arms for uniform, $p(1-p)$, and u_N sampling under the practical MaxRL estimator, no hindsight, and eight paired seeds. Checkpoints share a completion clock. The post-guidance analysis is descriptive.

Digits factorial (internally frozen negative). On a stored 1,077/360/360 train/development/test split of sklearn Digits, a 4,810-parameter float64 policy recomputes each example's exact correct-class probability before every update. We crossed practical MaxRL and RLOO with uniform, $p(1-p)$, and u_8 sampling. Five-rate development selected learning rate .1 for MaxRL, RLOO, and the common-rate check. Across 24 fresh paired blocks, the internally frozen interaction was $+.01589$ (paired bootstrap 95% CI $[-.01686, +.04712]$, exact two-sided sign-flip $p = .350$; 15/24 positive), so it was not supported despite passing the treatment-delivery gate. MaxRL did favor u_8 over

$p(1-p)$ by $+.20842$ (CI $[+.16791, +.24744]$; 23/24), but RLOO's frozen prediction reversed: $p(1-p) - u_8 = -.17665$ (CI $[-.22042, -.13593]$; 0/24). Moreover, the matched samplers were below uniform for both MaxRL ($-.11279$) and RLOO ($-.37581$). Because every selected rate equaled .1, the tuned and common schedules have byte-identical ledgers and checkpoints and identical scientific summary content after removing phase/authorization labels; they are identity checks, not independent replications. This exact-p contextual bandit therefore supports neither the planned interaction nor a universal curriculum advantage; it shows why expected coefficient mass must remain an estimator-activity proxy rather than a learning law. Its internally hashed local lock is not backed by an immutable public pre-execution commit that independently establishes timing.

Acrobot V3 (historical). The frozen control uses official Acrobot-v1, eight nested thresholds from -1.5 through native success at 1.0, one shared H64 actor, practical MaxRL, $N = 16$, and no hindsight. Uniform and u_{16} receive 20 paired seeds and a nominal two million actual transitions per run, completing the final group. Evaluation uses 32 shared trajectories total, reused across all eight predicates, at initialization and each 100,000 transitions with fixed common random numbers. The internally frozen primary is target-uniform normalized transition-AUC; group diagnostics and evaluation curves for all 40 runs are vendored. A post-hoc audit found cross-seed overlap between actor-initialization and action-sampling RNG roots, so the retained effect and mechanism audit are descriptive rather than clean confirmation. The mechanism audit is deterministic from those records.

Acrobot fixed-pool tournament V2. The replacement reruns uniform and u_{16} and adds $p(1-p) = u_2$ on fresh logical seeds 20000–20019. All 60 source/runtime-locked runs completed after an outcome-blind development gate; globally disjoint RNG roots and per-run group-diagnostic and aggregate-checkpoint ledgers were retained. The primary $u_{16} - p(1-p)$ transition-AUC difference is $+.04803$ (paired bootstrap 95% CI $[+.02094, +.07385]$, exact sign-flip $p = .003361$), clearing the predeclared $+.01$ estimate filter; Table 1 gives the Holm-controlled uniform comparisons. Observed mass means are close between adaptive arms (.7344 versus .7490 per group; 115.57 versus 115.90 per million transitions), both descriptive paired intervals include zero, and no equivalence test was frozen. The p-value tests zero rather than the SESOI, and pairing does not guarantee sign exchangeability. The $p(1-p)$ arm isolates only the score on our scaffold, not full ProCuRL or SFL.

Paid-probe ProCuRL selection semantics. This is a separate 80-pair family, not an extension of the V2 table. It keeps the same eight thresholds, H64 actor, practical MaxRL estimator, $N = 16$, and 3×10^{-4} learning rate. Its four arms are ordinary uniform; a uniform sham with paid probes; source-faithful ProCuRL selection $\mathrm{softmax}\{20\hat{p}(1-\hat{p})\}$; and continuous-range-matched u_{16} selection. Probed arms run 20 episodes per task initially and at every crossed 5,120-student-transition boundary; all student and probe transitions count toward the nominal two-million paid budget. Across seeds 21000–21079, the internally frozen fixed-paid-AUC contrast u_{16}−ProCuRL is $+.004894$ (paired SD .022139, $t_{79} = 1.9773$, $p = .05149$; 20,000-resample CI $[+.000110, +.009727]$; one-million sign-flip $p = .05161$). It misses both the .02 SESOI and the .05 paired-t criterion. The five Holm-family contrasts are ProCuRL–sham $-.001704$ (adjusted $p = .4568$), u_{16}–sham $+.003190$ (.4547), ProCuRL–ordinary $-.313774$ (3.47×10^{-66}), u_{16}–ordinary $-.308880$ (9.48×10^{-67}), and sham–ordinary $-.312070$ (4.06×10^{-68}). Mean fixed-paid AUCs are .33771 (ProCuRL), .33942 (sham), .65149 (ordinary), and .34261 (u_{16}).

The probed arms average 28.0–28.1 sweeps and 140.5–141.1k student transitions versus 2.003M for ordinary uniform; 93.19–93.21% of their paid transitions are probes. They average 15.4–19.8 optimizer updates versus 259.1 ordinary-uniform updates, with mean paid-budget overshoot 69,960–72,939 transitions versus 2,984. All 320 confirmation and 12 outcome-blind development runs reconcile exactly, all 11 development gates pass, and the common-random-number invariants replay.

An earlier outcome-blind development wave was invalidated before any gate or arm contrast because 73 entropy-sum replays differed by at most 7.28×10^{-12}. Its seeds were burned and its incident receipt is retained; the reported family uses fresh seeds and a replacement lock.

The external confirmatory raw is 1,374,886,097 bytes. Its SHA-256 boundary is:

raw: b1f8756c249effab8c77101c8bca73ddf708a5e143c18fe8742fd5712fdd7c12
lock: b7c7f76f6aaffa1fe65557717bfe545f2ec850495d370cd97fe72a8871fc8d0f
analysis: 2010e30b5b15a212e2d6bdfaacd43d2434e5f468a96be613cf59744b9bc2fb38

The source bundle carries a 320-record external-raw manifest: compact mode checks its small-artifact bindings, while full verification requires the raw file and replays locked validation. The local lock is mechanical provenance, not an immutable public pre-execution registration.

Post-guidance rollout-allocation factorial. We also compared fixed $N = 16$, published-HORA-style posterior hit allocation (Wang et al., 2026), and a mass-aware marginal that counts four already-spent probe rollouts. A 2-sampler by 3-allocator CPU factorial uses 16 paired seeds and 51,200 completions per cell. The mass-aware arm improves pass@8 AUC over HORA-style allocation by $+.00676$ (11/16 positive pairs) and over fixed groups by $+.02517$ (14/16). However, it reduces realized coefficient mass per completion by .000306 relative to HORA-style allocation in all 16 pairs, falsifying the frozen mechanism prediction. Its group sizes are also extreme (cell-average maximum about 76 versus 57 for HORA and 16 fixed). We therefore treat the result as exploratory motivation for posterior-calibration and allocation-cap ablations, not a fourth contribution.

A frozen follow-up crossed two samplers, the two adaptive scores, four caps, and three information sources over 16 paired seeds (50 cells; 800 runs). Restricting to the 32 deployable-information adaptive cells, all 32 adaptive-minus-fixed mean pass@8-AUC contrasts were positive, with 22 unadjusted descriptive Student-t intervals above zero; no multiplicity-adjusted inference is assigned. Yet realized coefficient mass was lower in 28/32 cells, with 24 intervals below zero and none above zero. The remaining 16 oracle-preupdate cells are diagnostic. Thus the expanded result reinforces the learning–mass mismatch rather than a mediation claim. The frozen engineering rule chose cap 32: averaged over samplers for the fresh-group/same-step cells, it reduced maximum group size by 58.07% with an AUC change of $-.00271$ versus uncapped. This is a point-estimate filter for a future pilot, not an equivalence test. The source/runtime lock is internally hashed but lacks a public pre-execution commit establishing timing.

Maze factorials. Wave 1 has six shared seed/warm-start blocks and six arms per block: MaxRL and GRPO under uniform and FrontierMax sampling plus two diagnostic GRPO arms. Its recorded endpoint hypothesis failed. Wave 2 has six fresh blocks and the four core arms. The external execution record names time-integrated pass@8 relative to warm start primary, with at least 5/6 positive blocks required under each sampler. The block analyzer reports sampler-specific exact sign tests, sampler-averaged contrasts, post-hoc intervals, and leave-one-block-out checks. No pooled p-value treats sampler observations as independent.

AUC robustness status. The frozen multiverse covers integration rule, warmup inclusion, horizons, and leave-one-checkpoint-out variants, but all 24 wave-2 checkpoint JSONLs and warm starts remain external. Structured summaries recover the anchors (uniform 6/6, mean $+.01496$; FrontierMax 6/6, $+.02404$; sampler-averaged 6/6, $+.01950$), but cannot identify those variants; the analyzer therefore exits explicitly blocked.

Countdown. We fine-tune SmolLM2-360M with $N = 16$. The v2 pool uses operand permutation, parenthesization, and exact division. Tier 1 has 128 evaluation tasks and 16 verifier-scored samples per task. B1 disables recycling; B2 admits exact-verifier achieved goals; the committed scoreboard contains three-seed aggregates. ARM B disables recycling and uses two optimizer epochs on every live group; its three individual endpoints are vendored. Headline endpoints are step-60 mean@16 and VERL bootstrap best@16. The latter uses 1,000 with-replacement resamples of size 16 per prompt and is not standard unbiased pass@16; missing per-task outcomes prevent recomputation. A stored task-identity summary reports zero tier-1 and tier-2 SFT overlap and 27 exposed tier-0 tasks; the source manifests needed to recompute those counts are absent. Tier 0 is excluded from absolute claims.

The reported recycling-package aggregate moves tier-1 mean@16 from .278 ± .066 to .324 ± .014 while bootstrap best@16 moves from .541±.024 to .492±.013 (mean ± sample SD across three seeds). Complete three-seed B1/B2 records are not vendored, so this supports neither a paired seed test nor a named-method effect claim. It is retained only as a measurement caution: mean accuracy alone cannot reveal whether coverage was spent to buy it.

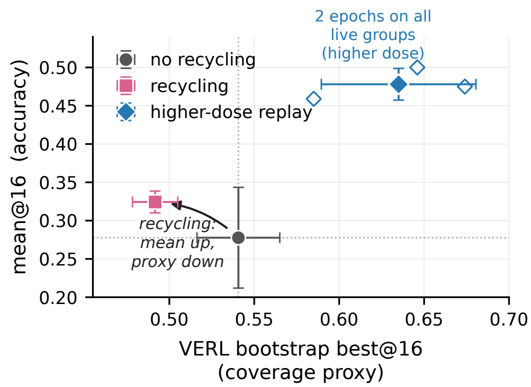

Figure 5: Secondary Countdown accuracy–coverage-proxy record. Error bars are sample standard deviations across three seeds; seed 3 for B1/B2 is aggregate-only. The recycling-package aggregate has higher mean@16 and lower VERL bootstrap best@16. The live-group replay control uses two optimizer epochs on every live group, so it cannot separate dose from update direction.

The externally specified replay control reaches mean@16 .478 ± .021 and bootstrap best@16 .635 ± .046, with all three endpoints vendored. Recycling permits at most 12 auxiliary groups per 64 requested groups (18.75%; realized dose can be lower), whereas replay doubles optimizer epochs on every live group. This higher-dose alternative improves both logged metrics without identifying a dose-monotonic causal bound.

Negative branches. The artifact retains the failed maze endpoint, gate, Jugs, and robotics branches. GSM8K missed its delivery threshold (.601480 versus $<$ .60) and is inconclusive; the single-seed Countdown per-k curve is exploratory. The corrected-code gate dose–response (frozen protocol, three paired seeds, independent A100 environment, retained 16-bit raw outcomes per task) passed its transfer gate — ungated recycling beat no-recycling on mean@16 in 3/3 seeds — and its manipulation check (rejection graded 0/.72–.80/.88–.90/.93 across gate strengths), but neither tested setting met the frozen usefulness criterion; per-seed signs were mixed at $n = 3$. Its raw outcomes also record, descriptively, one seed in which mean@16 rose while standard observed-set pass@16 fell .656 → .414 — a non-proxy instance of the ambiguity the operational recommendation targets.

D Artifact and data accounting

bash reproduce.sh –build tests the identities and integrations, verifies declared manifest inputs and the local compact registry, runs stored endpoint spot checks, regenerates figures, and builds the canonical main_iclr2027.pdf plus the extended main.pdf. The compact registry has 53 records: 35 maze, 11 Countdown, and 7 GSM8K. The broader 562-row registry used during release-branch auditing is a different object; it is not copied over this checkout's registry and is not used for the local count.

Missing external data are maze checkpoints; Countdown task manifests, outcomes, and B1/B2 seed 3, so the clean 101-task tier-0 analysis cannot be reconstructed; the 1.375 GB paid-probe confirmatory raw (its content hash and per-run receipt manifest ship); and the Digits 5.08 GB replay payload (block contrasts and a content manifest ship). Compact inputs and checksums are in paper/results/manifest.json and fail closed.